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Multi-type and multi-scale geological three-dimensional modeling using entity-relationship networks

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ABSTRACT

Three-dimensional geological models (3DGE) are essential for regional geological surveys, resource exploration, and engineering hazard assessment due to their ability to visualize subsurface stratigraphy, structural frameworks, and geological chronology. However, the accuracy of traditional modeling is constrained by borehole density and spatial distribution, making 3D geological reconstruction in areas with sparse or no borehole data particularly challenging. This study proposes an engineering-oriented methodology for constructing multi-type and multi-scale 3DGE using an entity-relationship network. The objective of this study is to develop a knowledge-driven 3DGE framework that can effectively reconstruct subsurface geological structures under absent real boreholes. By using large language models under the constraint of prompt templates to extract information from geological texts, a deep learning algorithm is then used to identify the spatial information in geological maps. It is organized into an expandable entity-relation network, including attributes such as geological ages and stratigraphic categories. Based on this network, virtual boreholes are generated by sampling along the inferred stratigraphic sequences, and the orientations and inclinations of the strata are converted into vertical constraints for modeling, enabling the flexible construction of three-dimensional models while meeting engineering requirements and balancing computational costs. Experimental results demonstrate that the proposed framework can effectively integrate textual and graphical information to construct 3DGE. The consistency rate between the constructed model's profile and the real profile reached 86.71%. This paper provides an efficient, scalable solution for regional 3DGE in areas with limited borehole resources, and offers a new technological approach for converting heterogeneous geological knowledge into underground 3DGE. It can be widely applied in resource exploration, engineering design, and geological hazard assessment.

1. Introduction

In geological surveys and regional modeling research, three-dimensional geological models serve as a crucial link between surface observations and the understanding of subsurface structures. Compared with one-dimensional textual descriptions and two-dimensional geological maps, three-dimensional models provide intuitive representations of stratigraphic distribution, structural patterns, and geological evolution. They enhance the understanding and prediction of subsurface architectures in structurally complex regions and support resource

exploration, engineering planning, and hazard mitigation [1–3]. Additionally, three-dimensional geological models have found widespread applications across multiple domains, including hydrogeology for groundwater flow and contaminant transport simulation [4], environmental geoscience for subsurface contamination assessment and remediation planning [5], and engineering geology for infrastructure planning, tunnel design, and slope stability analysis [6]. With the growing scale of engineering activities and the increasing need for timely decision-making, modeling objectives have expanded from site-specific or local engineering scenarios to large-area and multi-scale

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regional modeling. However, data sparsity and spatial heterogeneity at large spatial extents significantly limit model reliability and engineering applicability. On one hand, boreholes and borehole-like data provide unmatched vertical accuracy, yet their deployment is often restricted by time, cost, and terrain conditions, making comprehensive coverage difficult to achieve [7,8]. On the other hand, the demand for rapid acquisition of reliable geological information often conflicts with the long modeling cycles required by traditional methods. This contradiction becomes more pronounced in engineering scenarios that require fast and robust decision-making [9,10].

Over the past decades, three-dimensional geological modeling has undergone substantial methodological development while expanding into a wide range of application domains. Early foundational studies established the mathematical and computational principles for constructing subsurface interfaces and volumetric representations from geological observations, providing the basis for modern three-dimensional modeling frameworks [11]. Building on these foundations, subsequent reviews systematically summarized advances in geological modeling, highlighting the integration of multi-source data, geostatistical frameworks, and uncertainty quantification strategies that support regional-scale modeling and the interpretation of complex geological systems [12]. As modeling objectives gradually shifted from local-scale visualization to regional-scale and engineering-oriented applications, increasing attention has been paid to the challenge of reconstructing geologically consistent three-dimensional models under heterogeneous data conditions and sparse sampling. Recent studies have emphasized the need for scalable and flexible modeling workflows that can accommodate limited subsurface observations while maintaining geological realism [13–15]. In this context, recent studies have focused on enhancing the construction of three-dimensional geological bodies under limited borehole conditions by integrating multiple-point statistics with data-driven approaches, including deep learning. Incorporating geological organizational information through prior models or training images [16]. Auxiliary data such as two-dimensional cross sections or prior models are frequently used to strengthen the coherence of local stratigraphic structures [17], while algorithmic improvements including localized scanning and optimized simulation paths reduce partitioning artifacts and improve computational efficiency [18]. For example, Guo et al. (2022) [19] emphasized the significance of geological constraints in prior models and constructed an initial model that preserves geological consistency. They further introduced a moving window strategy to account for spatial non-stationarity, successfully reducing discontinuities along partition boundaries. Hou et al. (2023) [20] applied deep learning to automatically identify key geological objects in two-dimensional cross sections and transformed the extracted semantic relationships into prior constraints for multiple-point statistics, which improved both recognition and reconstruction of complex structures. Wang et al. (2025) [21] combined principal component analysis with autocorrelation functions to characterize the spatial structure of thin ore bodies and proposed optimized simulation paths and refined scanning strategies.

In parallel with these statistical and learning-based strategies, several studies have explicitly explored the use of virtual or surrogate boreholes as borehole-like vertical constraints to support three-dimensional geological reconstruction under data-sparse conditions. For example, Guo (2024) [22] formulated virtual borehole generation as a data-driven prediction problem, employing stacking ensemble learning to infer stratigraphic boundary elevations and detailed stratigraphic sequences from limited borehole samples, thereby enabling the construction of virtual boreholes for subsequent three-dimensional modeling. In a different line of work, virtual boreholes were derived from expert-interpreted local stratifications and geophysics-informed stratigraphic knowledge, and then assimilated into probabilistic geological models to support underground digital-twin applications and large-scale engineering modeling (Guo et al., 2025) [23]. These approaches collectively improve spatial coherence and reduce

computational burdens. However, despite their advances, most existing methods, including MPS-based, learning-assisted, and virtual-borehole-based approaches, still rely heavily on high-quality borehole data, representative training samples, or manually encoded expert knowledge for constraint definition. This reliance fundamentally limits their applicability and potential for engineering deployment in regions where boreholes are sparse or unavailable.

In parallel, long-term geological surveys have accumulated a substantial volume of multi-modal geological information, including regional geological chronicles, integrated geological maps, survey reports, and cross-section diagrams. These materials contain rich knowledge on stratigraphic sequences, lithological associations, contact relationships, and structural topology, and represent valuable sources for constructing geologically consistent regional models [24–26], especially when borehole data are insufficient. Nevertheless, these multi-modal resources differ significantly in narrative format, naming conventions, and compilation period. Their semantic systems lack direct correspondence, which leads to difficulties in information fusion, including semantic heterogeneity, misalignment between geometry and semantics, and conflicts across modalities. As a result, such information cannot be directly used for modeling and instead requires a unified framework capable of extracting, aligning, and resolving inconsistencies to form a structured geological knowledge base suitable for three-dimensional modeling.

Methods for extracting textual geological information have evolved from rule-based and dictionary-based approaches [27], to machine-learning-based entity and relation extraction [28], and further to semantic parsing enabled by deep learning and pre-trained language models [29]. Models such as BERT-BiGRU-CRF [30] and BERT-BiGRU-Attention-CRF [31] have significantly improved the accuracy of parsing complex lithological compositions, stratigraphic age expressions, and structural descriptions. More recently, prompt-based learning with large language models (LLMs) has enabled unified semantic extraction across tasks, demonstrating strong capability for handling complex geological terminology even under zero-shot or few-shot settings [32–34]. For map information extraction, research has progressed from conventional symbol recognition and semi-manual vectorization to deep-learning-driven semantic segmentation and object detection systems such as U-Net, DeepLab, Faster R-CNN, and YOLO. These techniques have been increasingly applied to lithology segmentation, fault detection, and boundary extraction [35–37]. However, most existing methods focus on single-modality processing and lack systematic integration of semantic information, geometric structures, and geological rules. **Cross-modal conflict detection, information completion, and consistency correction are still insufficient, and no complete knowledge chain has been established that can directly support three-dimensional modeling.**

To overcome these limitations, the objective of this study is to develop a knowledge-driven regional three-dimensional geological modeling framework that can reconstruct geologically consistent subsurface structures under sparse or even absent borehole conditions. **The proposed regional modeling framework integrates geological map and text information, aligns heterogeneous knowledge, generates virtual boreholes, and constructs three-dimensional models.**

The methodological novelty of this study lies in transforming heterogeneous geological knowledge into modeling-ready spatial constraints through a unified entity-relationship framework. By jointly integrating deep-learning-based geometric parsing and semantic extraction from pre-trained language models, stratigraphic attributes from texts and spatial, topological information from geological maps are systematically structured to generate virtual boreholes, providing effective subsurface constraints in regions lacking real drilling data.

Based on these virtual boreholes, a multi-scale and multi-type three-dimensional geological modeling approach is further developed. Through flexible use of virtual borehole constraints at a range of spatial resolutions (e.g., 10 m, 50 m, and 100 m), the proposed approach

supports unified modeling of stratigraphic structures, geological ages, and other types of three-dimensional geological information. This provides a novel technical pathway for building large-scale engineering models in regions where borehole data are sparse. The intended application scope of this framework includes regional-scale geological modeling in data-limited settings, with particular relevance to resource exploration, engineering geological analysis, and geological hazard assessment.

The structure of the paper is as follows: Chapter 2 introduces the methods for information extraction and three-dimensional model construction. Chapter 3 describes the construction of the experimental dataset and presents the experimental results. Chapter 4 introduces the innovation points of this paper and compares them with the existing methods for constructing three-dimensional models. Chapter 5 concludes the study.

2. Methodology

2.1. Overall framework

The overall framework of this study is illustrated in Fig. 1. It consists of four major components: multi-modal geological information extraction, knowledge graph fusion and correction, structural organization of stratigraphic information, and construction of three-dimensional

geological models based on spatial interpolation and spatial constraints. First, in the stage of multi-modal geological information extraction, **two parallel information processing workflows** are implemented for geological maps and geological texts. A **textual information extraction model** is developed using LLMs and prompt-based templates. Concurrently, deep-learning-based algorithms are employed to segment geological maps and extract key spatial features, followed by the establishment of a transformation between planar coordinates and geographic coordinates. Second, a unified framework for **information conflict detection and resolution** is constructed, which integrates extracted text and map information into a coherent multi-modal knowledge graph. This step produces a coherent **multi-modal fused knowledge graph** that ensures semantic and topological consistency across different geological information sources. Third, stratigraphic reasoning based on the knowledge graph is performed **to derive virtual borehole data**. This includes lithological sequences, geological ages, layer thicknesses, dip directions, and dip angles. These virtual boreholes provide essential vertical constraints in regions where real drilling data are sparse or absent. In addition, multi-scale surface constraints are generated through geological map rasterization and three-dimensional DEM filling, providing lateral and topographic constraints for three-dimensional modeling. Finally, spatial interpolation algorithms are applied to the virtual borehole data to generate geological models at spatial resolutions of ten meters, fifty meters, and one hundred meters.

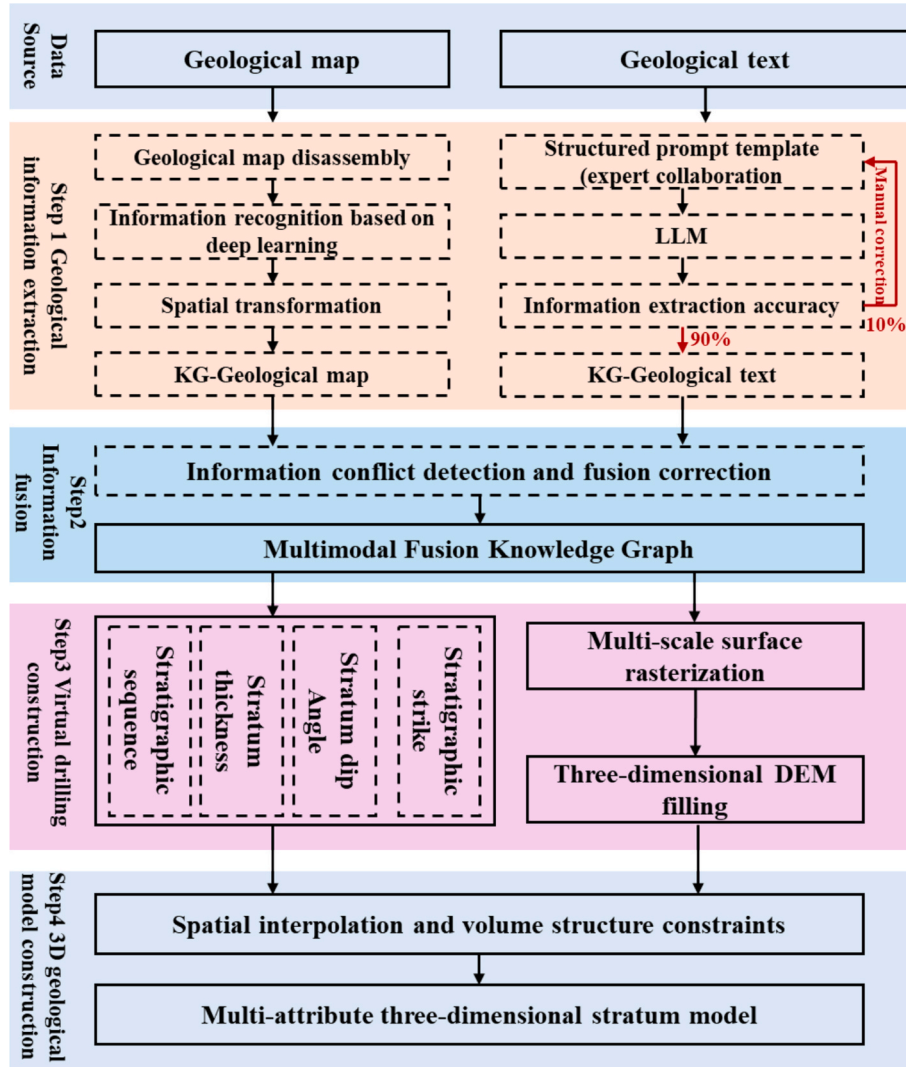


Fig. 1. Overall framework.

This multi-resolution modeling enables efficient adaptation to different application requirements in engineering geology and regional-scale analysis.

2.2. Extraction and alignment of textual and graphical information

2.2.1. Geological text information extraction using prompt-based LLM

To enhance the capability of LLM in understanding structured geological knowledge and identifying semantic relationships, this study proposes a prompt-engineering extraction method that incorporates structured knowledge tables. The central idea is to use these structured tables as external prompts that guide the model to recognize entities and their attribute relationships within geological texts, thereby aligning tabular knowledge with textual semantics.

During the model training stage, each training instance consists of a text-table pair (T_i, G_i) together with the corresponding entity-relation label y_i . Through prompt-based templates, the model learns implicit relational patterns between geological entities and their attributes. The prompt template is generated by transforming the structured table G_i into a natural-language description P_G , which is then concatenated with the corresponding text T_i to form the final input:

$$x_i = \text{Template}(P_G, T_i) = [\text{CLS}]P_G[\text{SEP}]T_i[\text{SEP}] \quad (1)$$

The input is processed by a pretrained Transformer encoder to obtain the global contextual representation of the text-prompt pair:

$$h_i = \phi_{\text{enc}}(x_i) \quad (2)$$

The contextual vector h_i is then mapped to the relation-label space through a classifier f_θ , yielding the probability distribution of entity relations:

$$\hat{y}_i = f_\theta(h_i) = \text{softmax}(W \bullet h_i + b) \quad (3)$$

where W and b are the classifier parameters.

The model is optimized by minimizing the cross-entropy loss:

$$L = - \sum_i \log p(y_i | x_i) \quad (4)$$

During the testing stage, for a new text T , a structured table G with the highest similarity is selected to generate a prompt P_G . The final input is constructed as:

$$x' = \text{Template}(P_{\{G^*\}}, T) \quad (5)$$

and the model predicts the potential entity relations contained in the text.

To improve model generalization and accuracy, a closed-loop refinement mechanism is adopted. When the confidence of predicted relations falls below a predefined threshold (this paper set threshold is 90%), a small subset of samples is randomly selected for manual verification and iterative correction. This loop continues until the model achieves a stable accuracy level. The prompt-based method retains the advantages of conventional relation extraction while emphasizing the incorporation of structured knowledge prompts and iterative refinement. It ensures high-precision extraction of entity relations under complex geological descriptions and provides reliable inputs for subsequent virtual borehole generation.

2.2.2. Extraction of geological map information

A comprehensive geological map consists of multiple components, including the main map, stratigraphic column, cross-section, and legend. These components contain layered geological information ranging from stratigraphic distribution and contact relationships to vertical sequences and structural patterns. Due to the complexity of these maps, a unified extraction framework is required. This study proposes an integrated map-parsing framework consisting of three principal steps: map decomposition, legend structuring, and spatial

interpretation of the main map. The relationships among these components are shown in Fig. 2.

(1) Map Decomposition and Structural Identification

Considering the standardized internal structure of geological maps and stratigraphic columns, Mask R-CNN is adopted as a unified instance-segmentation model to automatically identify and decompose individual map components. By applying pixel-level annotations to training samples [38–40], the model predicts masks corresponding to the main map region, legend region, section region, and scale bar.

Mask R-CNN uses a ResNet-18 backbone with a feature pyramid network to extract multi-resolution features. Region-of-interest alignment is used to obtain precise regional features, which are passed to the mask branch to generate component regions:

$$M_i = \{(x, y) | (x, y) \in \text{component}_i\} \quad (6)$$

where i denotes the component index (for example, main map, stratigraphic column, legend). (x, y) represents the pixel coordinates of the image.

For stratigraphic columns, table-structure parsing is performed after segmentation to obtain structured layer identifiers, lithological descriptions, and thickness information.

(2) Construction of the Legend “Color-Text” Database

Each legend entry consists of a paired color patch and text description. This study constructs a “color-text” database D through color-patch detection, text recognition, and normalization.

For each color region R_i^{color} , the mean RGB values of the included pixels are computed to derive a unique color vector:

$$C_i = \left[\frac{1}{m} \sum_{k=1}^m R_k, \frac{1}{m} \sum_{k=1}^m G_k, \frac{1}{m} \sum_{k=1}^m B_k \right] \in R^3 \quad (7)$$

where m is the number of pixels in the region, and R_k, G_k, B_k are the RGB values of pixel k .

Optical character recognition (OCR) is applied to obtain the corresponding text T_i , which is normalized using a geological dictionary. The final legend database is defined as:

$$D = \{(C_i, T_i) | i = 1, 2, \dots, N\} \quad (8)$$

where N is the number of legend entries.

(3) Spatial Information Extraction and Vectorization of the Main Map

Based on the legend database, a complete workflow is designed to convert the main map from pixel space to geographic coordinate space. The process includes pixel classification, region aggregation, boundary vectorization, and coordinate transformation.

The main image pixel $I(x, y)$ is assigned a geological unit label through Euclidean-distance matching with color vectors C_i :

$$\text{Label}(p) = \arg\min_i \|I(x, y) - C_i\|_2 \quad (9)$$

This produces a pseudo label image $L = Z^{H \times W}$ in which every pixel is assigned a unit category. Connected regions are then extracted:

$$A_i = \{(x, y) | L(x, y) = i, (x, y) \in \Omega\} \quad (10)$$

where A_i is the set of pixels corresponding to unit i , and Ω is the domain of the image.

Boundary extraction from each region yields a closed polygon in pixel-coordinate space:

$$P_i^{\text{flat}} = \text{Contour}(A_i) = \{(x_1, y_1), \dots, (x_n, y_n)\} \quad (11)$$

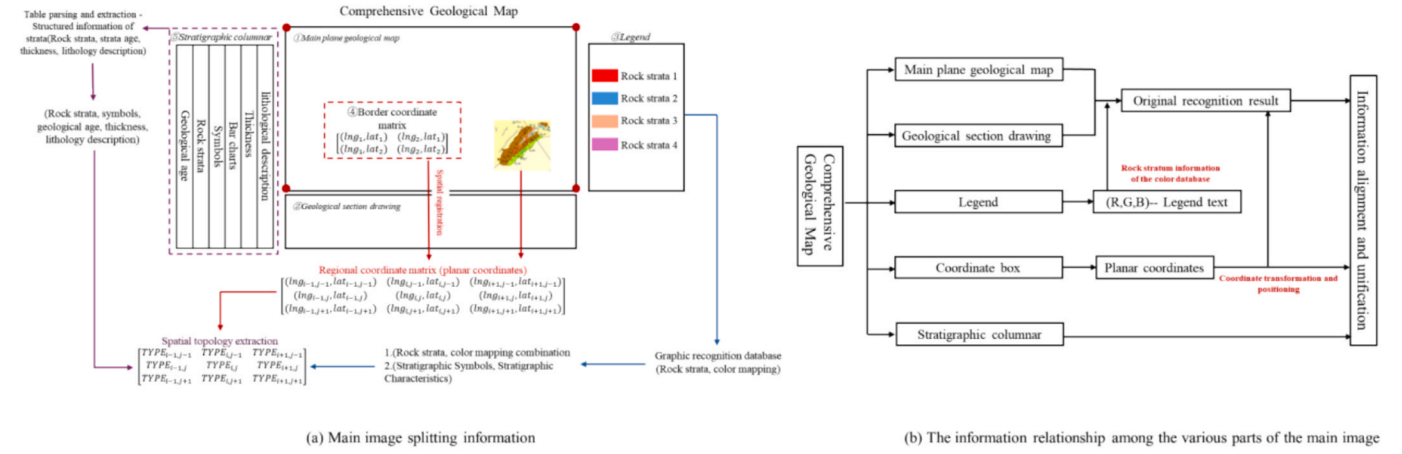


Fig. 2. Extraction of comprehensive geological map information.

where n is the number of boundary points.

Finally, an affine transformation matrix is estimated using the corner points of the map sheet to convert pixel coordinates to geographic coordinates:

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} a & b & t_x \\ c & d & t_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (12)$$

The geographic-coordinate polygon is thus obtained as:

$$P_i^{geo} = \tau(P_i^{flat}) \quad (13)$$

The final output is a set of GeoJSON objects that encode both geometric boundaries and semantic attributes, such as lithology, geological age, and structural type—thereby enabling fully structured and spatialized representation of complex geological maps.

2.2.3. Information fusion and conflict correction

Structured knowledge extracted from texts and images often presents inconsistent descriptions of the same geological entity due to differences in narrative format, naming conventions, and compilation periods. These discrepancies lead to semantic heterogeneity, misalignment between geometric and semantic information, and conflicts across modalities. To unify the heterogeneous attributes obtained from textual extraction and map extraction, this study establishes an attribute alignment and conflict-correction workflow, as illustrated in Fig. 3.

For any geological entity, the extracted attribute sets from texts and images can be expressed as:

$$I_i = \{(A_k, V_k^t, p_k^t)\} \quad (14)$$

$$I_i = \{(A_k, V_k^i, p_k^i)\} \quad (15)$$

where A_k is the attribute field, V_k^t and V_k^i are the corresponding field values extracted from texts and images, p_k^t and p_k^i represent their confidence scores.

After attribute alignment, a comparison unit is constructed as:

$$U_k = (A_k, V_k^t, V_k^i) \quad (16)$$

To ensure consistent evaluation across different field types, several similarity measures are introduced, as summarized in Table 1.

A unified fusion strategy is constructed through a confidence-weighted consistency score:

$$S_k = w_t p_k^t \text{sim}_k(V_k^t, V_k^i) + w_i p_k^i \text{sim}_k(V_k^i, V_k^t) \quad (17)$$

where w_t and w_i are the weights assigned to textual and image sources, sim_k denotes the similarity function selected according to the field type.

Consistency decisions are made based on score thresholds: If

Table 1
Consistency evaluation methods.

Field type	Similarity formula
String fields	$\text{sim}_{str}(s, t) = 1 - \frac{\text{Lev}(s, t)}{\max(s , t)}$
Set fields	$\text{sim}_{set}(A, B) = J(A, B) = \frac{ A \cap B }{ A \cup B }$
Numerical fields	$\text{sim}_{num} = 1 - \frac{ v^t - v^i }{\max(v^t , v^i) + \epsilon}$
Relational fields	$\text{sim}_{rel}(v^t, v^i) \in \{-1, 0, 1\}$

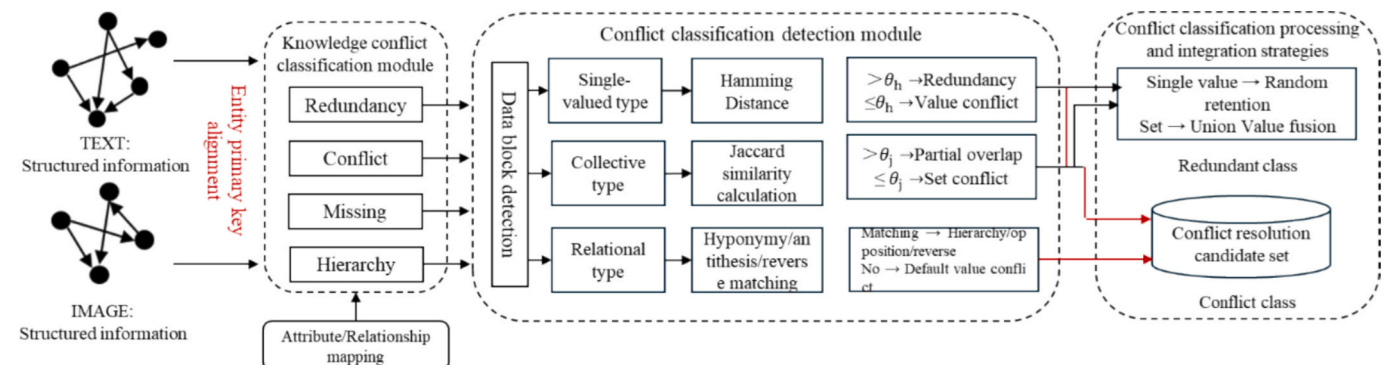


Fig. 3. Information fusion and conflict correction framework.

$S_k \geq \tau_{acc}$, the two values are considered consistent and directly fused; If $S_k \leq \tau_{flag}$, the pair is recorded as a conflict and added to the conflict set C ; Otherwise, a confidence-prior correction strategy is applied. The fused attribute value is computed through confidence-weighted averaging:

$$V_k^f = \frac{p_k^i V_k^i + p_k^j V_k^j}{p_k^i + p_k^j} \quad (18)$$

This workflow enables field-level consistency checking, conflict identification, and numerical fusion between map-derived and text-derived attributes. It ensures that the structured geological properties supporting subsequent modeling remain semantically unified and logically coherent.

2.3. Three-dimensional geological modeling based on the entity-relationship network

2.3.1. Generation of virtual boreholes

The construction of virtual boreholes relies on stratigraphic attributes stored in the knowledge graph, including thickness, lithology, strike, dip, and overlying and underlying relationships. Through Cypher queries, constraint information for target stratigraphic units can be retrieved from the heterogeneous graph database. The basic query structure is:

$$Q = \text{MATCH}(f : \text{Formation}) - [r : \text{HAS_PROPERTY}] - > (a : \text{Attribute}) \text{RETURN } f.\text{name}, a.\text{key}, a.\text{value} \quad (19)$$

where f represents a formation node and a represents an attribute node. The output consists of the attribute set associated with each stratigraphic unit.

Based on the query results, a stratigraphic attribute dictionary is constructed:

$$D = f_j : (\text{thickness}_j, \text{lithology}_j, \text{strike}_j, \text{dip}_j, \text{overlying}_j, \text{underlying}_j), j = 1, 2, \dots, N \quad (20)$$

Using digital elevation model data (DEM), the top and bottom surfaces of each formation unit are projected into space. Stratigraphic units are arranged according to their overlying and underlying relationships, forming a virtual borehole sequence. A virtual borehole can be expressed as:

$$B = \{(z_0, f_0), (z_1, f_1), (z_2, f_2), \dots, (z_m, f_m)\} \quad (21)$$

where $z_0 > z_1 > \dots > z_m$ represents the elevation sequence of formation boundaries, and f_m is the corresponding stratigraphic unit. The virtual borehole records a complete vertical sequence from the surface downward, providing essential input data for subsequent modeling.

2.3.2. Construction of the three-dimensional geological model

Based on the generated virtual boreholes, a spatial interpolation algorithm is developed for three-dimensional geological modeling. By measuring the spatial distance between interpolation points and known sample points, and by incorporating local spatial correlation, the method reconstructs geological structures through attribute prediction and voxel filling.

(1) Planar Rasterization Based on the Vectorized Geological Map

From the vectorized main geological map, the set of surface polygons representing formation distribution is obtained:

$$M = \{P_j^{\text{surf}} | j = 1, 2, \dots, N\} \quad (22)$$

where P_j^{surf} denotes the surface boundary of formation j .

These polygons are converted into a two-dimensional raster:

$$G_{xy} = \{(x_p, y_p, l_{pq}), l_{pq} \in \{f_1, f_2, \dots, f_N\}\} \quad (23)$$

where (x_p, y_p) is the grid point and l_{pq} is the stratigraphic label at the surface.

(2) Vertical Filling Based on Virtual Boreholes

For each formation unit f_j , thickness as well as overlying and underlying relations are derived from the virtual borehole sequence B . At each surface grid point (x_p, y_p) with surface formation label l_{pq} , vertical extension is applied according to dictionary D . This process yields spatial columns:

$$C_{pq} = \{x_p, y_p, [z_{top}^j, z_{bot}^j], f_j\}, j = 1, 2, \dots, m \quad (24)$$

where $[z_{top}^j, z_{bot}^j]$ represents the top and bottom elevation of formation f_j .

(3) Three-Dimensional Spatial Interpolation and Voxel Construction

By aggregating all columns C_{pq} , a three-dimensional point set is obtained:

$$P = \{(x_p, y_p, z, v)\} \quad (25)$$

where v is the corresponding stratigraphic label.

The spatial distribution of formations depends not only on neighborhood relations but also on geometric orientation. Therefore, strike-dip constraints are introduced. The orientation of formation f_j is represented by a normal vector:

$$n_j = (\sin\theta_j \cos\phi_j, \sin\theta_j \sin\phi_j, \cos\theta_j) \quad (26)$$

where θ_j is the dip angle and ϕ_j is the strike direction.

During interpolation, the spatial distance between an interpolation point (x, y, z) and an observed point (x_i, y_i, z_i) is modified by a directional weight:

$$d_{ij} = \|(x, y, z) - (x_i, y_i, z_i)\| + \lambda \cdot \left| n_j \cdot \frac{(x - x_i, y - y_i, z - z_i)}{\|(x - x_i, y - y_i, z - z_i)\|} \right| \quad (27)$$

where λ is the coefficient controlling the influence of formation orientation.

The final classification is determined by:

$$v(x, y, z) = v_i^*, i^* = \text{argmin}_i d_{ij} \quad (28)$$

This interpolation method incorporates both spatial proximity and geological orientation, resulting in voxel structures that more accurately reflect realistic geological configurations.

2.3.3. Accuracy assessment and evaluation

To evaluate the validity of the constructed three-dimensional model, this paper examines the model from the perspective of geometric consistency [41,42].

A reference geological section S_{ref} is extracted from the comprehensive geological map, and a corresponding model-derived section S_{model} is cut from the three-dimensional model.

$$S_{\text{ref}} = \{(x_i, z_i, v_i)\} \quad (29)$$

$$S_{\text{model}} = \{(x_i, z_i, v_i^{\text{model}})\} \quad (30)$$

where (x_i, z_i) is the coordinate of sampling point i , v_i is the true stratigraphic label, and v_i^{model} is the label predicted by the model.

The section consistency accuracy is computed as:

$$ACC = \frac{1}{M} \sum_{i=1}^M \mathbb{I}(v_i = v_i^{model}) \quad (31)$$

where M is the number of sampled points and $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the condition is satisfied and 0 otherwise.

3. Experimental results and analysis

3.1. Dataset

3.1.1. Dataset construction and preprocessing

The textual and graphical datasets used in this study were collected from the National Geological Archives of China (<http://www.ngac.org.cn>) and the Zhejiang Geological Data Online Service Platform (<https://dzsl.zjdz.org.cn>). The data formats and quantities are summarized in Table 2. Textual data were preprocessed through format conversion, segmentation, and removal of irrelevant content. All map data were converted to TIF format, and fifteen geological maps were annotated using LabelMe. The annotated components included the main geological map, geological section, scale bar, stratigraphic column, and legend.

3.1.2. Experimental platform

Experiments were conducted using Python 3.6 and Tensorflow 2.5. The hardware configuration consisted of an NVIDIA GeForce RTX 3070 GPU with 16 GB memory, a 2.90 GHz Intel Core i7-10700 CPU, and 16 GB RAM.

3.2. Results of text–image information extraction and alignment

3.2.1. Results of textual information extraction

To obtain structured knowledge related to Stratigraphic unit and geological profiles, prompt templates were designed by a team of six geological professionals (two geological experts and four field geologists). The template design was guided by common structures observed in geological descriptive texts, determining extraction categories and attribute fields. All sample texts were manually annotated to construct an entity–relation extraction dataset centered on Stratigraphic unit. After manual validation and correction, a total of 744 annotated entries were produced and divided into training, validation, and testing sets at a ratio of 8:1:1. The extraction categories and template contents are listed in Table 3.

In this study, a prompt-based fine-tuning strategy was adopted using GPT-4o (2024-11-20). It maintains a comparable intelligence level to GPT-4 Turbo, while achieving approximately twice the speed and a 50% improvement in cost-effectiveness. GPT-4o also demonstrates enhanced performance in handling non-English languages and visual tasks [43,44]. In contrast to conventional named entity recognition models, such as BERT-BiLSTM-CRF, which typically contain on the order of 10^8 parameters and rely on sequence labeling architectures, GPT-4o enables joint extraction of entities, attributes, and hierarchical relationships within a unified structured output. This capability is particularly suited to geological texts, where stratigraphic descriptions are often implicit,

multi-level, and highly context-dependent.

In model training, the Adam optimizer was employed with a learning rate of 1×10^{-4} , two training epochs, batch size of sixteen, LoRA dropout of 0.05, and a maximum gradient clipping threshold of 0.3. During prompt-based fine-tuning, only the loss between the model-generated outputs and the annotated targets was backpropagated, ensuring that gradient updates focused on the extraction task itself.

During training, the loss curves of the prompt-finetuned GPT-4o and the BERT-BiLSTM-CRF model are shown in Fig. 4(a). The prompt-finetuned GPT-4o achieved the lowest average loss (0.5373), with stable convergence, indicating its strong ability to learn structured extraction patterns. By contrast, the BERT-BiLSTM-CRF model produced a significantly higher loss (1.39), demonstrating clear limitations in handling generative structured extraction tasks.

For a systematic evaluation of entity–relation extraction performance, Precision (P), Recall (R), F1-score (F1), and inference time were adopted as metrics. The comparison models included prompt-finetuned GPT-4o, GPT-4o without prompt templates, and BERT-BiLSTM-CRF. As shown in Fig. 4(b), prompt-finetuned GPT-4o achieved the highest F1 score (93.3%), with precision of 94.0% and recall of 92.7%. These results demonstrate its superior performance in entity recognition, semantic understanding, and structural inference. In contrast, BERT-BiLSTM-CRF showed a significantly lower F1 score (73.2%) and the longest inference time (123.7 s). GPT-4o without prompt templates exhibited a large performance decline, with F1 score dropping from 93.3% to 74.5%. This confirms that prompt templates not only impose structural constraints but also significantly enhance the stability of semantic interpretation and domain terminology recognition in professional extraction tasks.

Table 4 summarizes the extraction accuracy of individual attribute fields. Without prompt templates, LLMs showed significantly weaker performance in fields requiring semantic reasoning, domain terminology interpretation, or complex descriptive understanding. The F1 scores for such fields, including “layer description,” “overlying formation,” and “restoration notes”, were generally below 50%. After integrating prompt templates, all these fields achieved F1 scores above 85%.

Similarly, numerical fields such as “longitude” and “thickness” showed an accuracy decrease of approximately 15% without prompts. These findings demonstrate that prompt templates substantially enhance structural constraints and domain-specific semantic guidance, enabling the large model to extract professional geological information more reliably. Therefore, the prompt-integrated GPT-4o model was selected as the core text-extraction module in this study.

3.2.2. Results of map information extraction

The structural recognition of geological maps was conducted using a unified Mask R-CNN framework, enabling automatic detection of the main map, geological sections, legend, coordinate frames, and tabular elements. To enhance robustness and improve detection accuracy, data augmentation was performed using rotation angles (30°, 60°, 90°). All components were trained jointly in a parallel training pipeline. A total of 1725 training samples were used, and twenty percent were randomly selected for five-fold cross-validation to assess model stability. The detection results, shown in Table 5, indicate that Mask R-CNN can reliably identify map components with high precision, providing a solid foundation for subsequent stratigraphic interpretation of the main geological map.

The main map serves as the core carrier of surface geological information, and its extraction depends on the legend-based “color–text” matching method, as illustrated in Fig. 5.

A combined strategy involving different color spaces (RGB, HSV, Lab) and classification methods (K-means, KNN, SVM) was evaluated to determine the optimal configuration for color-patch recognition. As shown in Table 6, the RGB color space combined with minimum Euclidean distance matching achieved the best overall performance, with a color-patch classification accuracy of 92.4% and text-mapping

Table 2

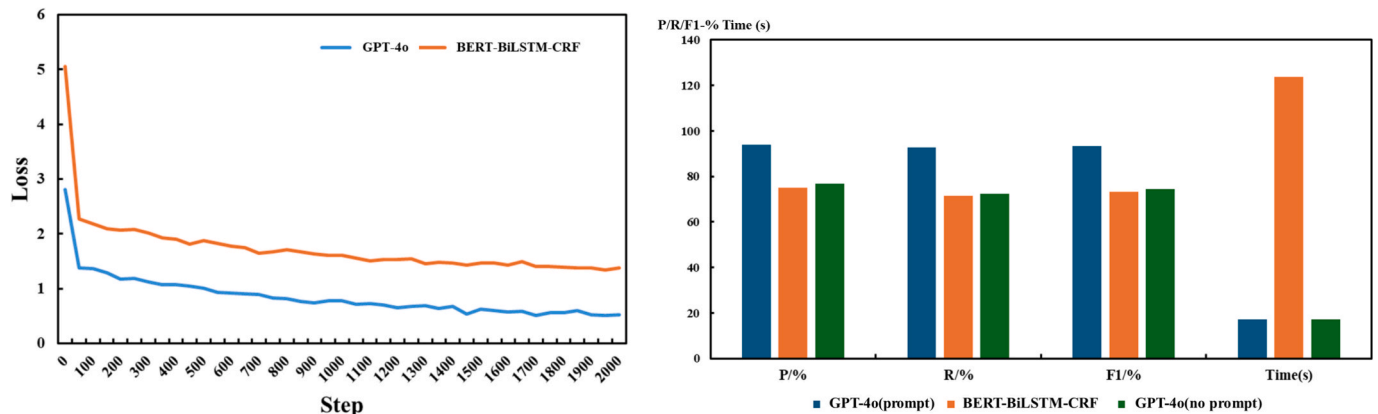
Dataset overview.

Data type	Source	Format	Quantity
Text	Regional geological survey reports	PDF	62
	Geological chronicles		1
Images	Standard geological maps	TIF	55
	Scale bars	PNG	100
	Stratigraphic columns	PNG	100
	Legends	PNG	100

Table 3

Categories and contents of prompt templates.

Extraction category	Extraction content					
Basic lithostratigraphic information	Stratigraphic unit	Formation code	Reference profile longitude	Reference profile latitude	Distribution	Lithological assemblage
Basic profile information	Geological profile	Stratigraphic unit	Location	Longitude	Latitude	Profile type
	Profile thickness	Overlying formation	Underlying formation			
Profile-layer information	Geological profile	Stratigraphic unit	Stratigraphic segment	Layer number	Layer description	Individual layer thickness

**Fig. 4.** Loss variations of different models.**Table 4**

Information extraction accuracy.

Extraction category	Extraction content	Prompt F1/%	No-prompt F1/%
Basic lithostratigraphic information	Stratigraphic unit	95.1	60.4
	Formation code	92.4	55.3
	Reference profile longitude	97.6	78.2
	Reference profile latitude	97.5	77.6
	Distribution	88.3	50.1
	Lithological assemblage	85	46.7
Basic profile information	Geological profile	94.2	58.6
	Stratigraphic unit	93.6	57.3
	Location	90.9	49.8
	Longitude	97.8	79.3
	Latitude	97.9	78.7
	Profile type	92.1	54
	Profile thickness	95	72.5
	Overlying formation	90.5	46.2
	Underlying formation	90.2	45.8
Profile-layer information	Geological profile	93.7	56.4
	Geological profile	91.8	55
	Stratigraphic segment	89.4	47.3
	Layer number	96	74.1
	Layer description	84.7	43.7
	Individual layer thickness	95.2	71.8

accuracy of 97.6%. This approach is simple, computationally efficient, and well suited for large-scale automated processing. Lab color space exhibited slightly better perceptual consistency but required more computational resources, while HSV suffered from class confusion in

low-saturation ranges. KNN and SVM showed advantages in fine-grained classification but required additional training datasets, making them less suitable for standardized workflows.

For the study area (Hangzhou, Zhejiang Province), a total of 160 representative geological-unit color patches were extracted from the comprehensive geological map. Their mean RGB values were used to construct the legend color database, enabling minimum-distance pixel matching across the main map. As shown in Fig. 6, different stratigraphic units were successfully labeled, achieving an overall pixel classification accuracy of 92.4%. Most misclassifications occurred near boundary regions where color similarity was high.

After color-based unit identification, all extracted geological-unit coordinates, initially in pixel space, were transformed into WGS 84 geographic coordinates using an affine transformation derived from sheet-corner registration. Using manually digitized boundaries as reference, the accuracy assessment (Table 7) indicates that the coordinate conversion achieved high spatial accuracy, sufficient for further geographic analysis. All geological units were automatically exported in GeoJSON format and stored in a database, completing the structured and spatially aligned representation of the comprehensive geological map.

3.2.3. Conflict detection and fusion of text–image information

Using the conflict detection and correction framework developed in this study, a total of 5716 geological entities extracted from textual materials and comprehensive geological maps of Zhejiang Province were evaluated. After field alignment and conflict assessment, the system identified 438 conflicting fields and 240 redundant fields. The distribution of different conflict and redundancy types is summarized in Table 8.

The analysis indicates clear imbalance across conflict types. String-type conflicts (e.g., stratigraphic names, color terms) are most frequent, primarily due to standardized or abbreviated naming in maps versus descriptive or region-specific terminology in textual reports. Lithological assemblage fields, as typical set-type attributes, often show

Table 5
Detection accuracy of map components.

Data category	Training samples	Detections	Missed detections	False detections	TPR	FPR	LPR
Main map	198	192	6	4	0.97	0.02	0.03
Profile	120	115	5	3	0.96	0.03	0.04
Legend	201	197	4	5	0.98	0.02	0.02
Coordinate frame	198	193	5	6	0.97	0.03	0.03
Table	210	205	5	7	0.98	0.03	0.02
Table header	210	206	4	8	0.98	0.04	0.02
Table content	210	207	3	9	0.99	0.04	0.01

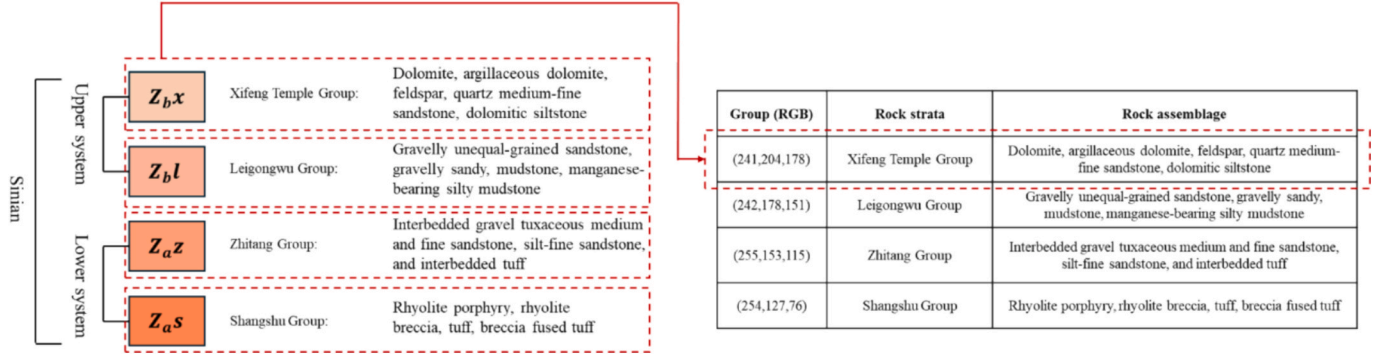


Fig. 5. Schematic diagram of text matching based on legends.

Table 6
Accuracy of legend information extraction.

Color space	PATCH accuracy (top-1)	Patch F1 mean	Text-mapping accuracy
RGB	92.4%	0.893	97.6%
HSV	89.1%	0.861	94.2%
Lab	91.3%	0.875	96.1%
RGB + KNN	94.2%	0.902	97.3%
RGB + SVM	93.6%	0.896	97.8%

granular mismatches such as “limestone with shale” versus separate items “limestone” and “shale,” resulting in set overlaps or redundancy. Relation-type conflicts (e.g., “unconformable contact,” “conformable

contact,” “tectonic emplacement”) account for about 13.1% and mainly arise because map extraction produces standardized terms, whereas textual descriptions often include hierarchical or oppositional expressions. Without domain-specific dictionary rules, these cases are easily misclassified as redundancy.

To ensure accurate conflict detection, similarity thresholds were

Table 7
Coordinate transformation accuracy.

Metric	Mean error	Maximum offset	Accuracy level
Longitude error	0.00042°	0.00093°	High
Latitude error	0.00037°	0.00088°	High
Spatial error (meters)	37.4 m	82.1 m	Acceptable

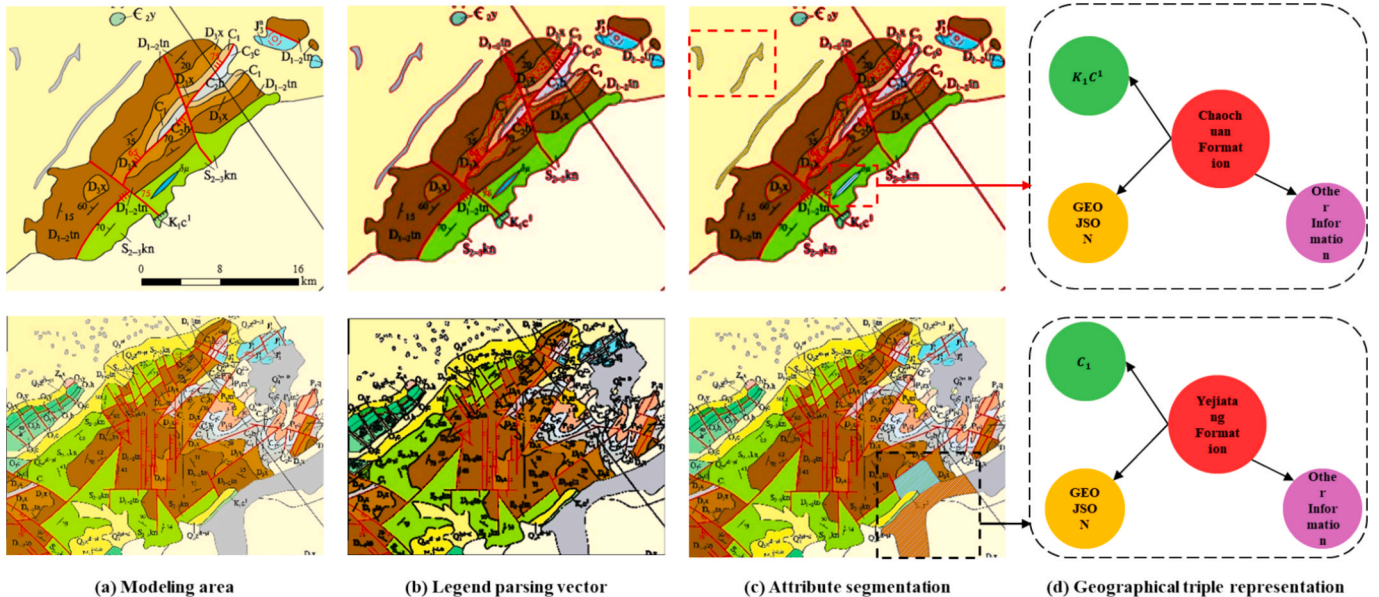


Fig. 6. Main image analysis and information extraction based on legends.

Table 8
Types of conflicts and descriptions.

Conflict type	Count	Ratio	Description
Value conflicts (string type)	213	34.4%	Differences in color codes, stratigraphic naming, or inconsistent abbreviations
Set-overlap conflicts	144	23.3%	Lithological assemblages extracted with different granularity in text and maps
Hypernym-hyponym conflicts	81	13.1%	Concentrated in “genetic relationship” and “contact relationship” fields
Redundant information (set type)	162	26.2%	Map-derived information more complete and contains all textual items
Redundant information (string type)	78	12.6%	Naming fields identical but format differs (Chinese vs English, parentheses, etc.)

applied: $\theta_h = 3$ for string fields and $\theta_j = 0.5$ for set-type fields. Under these thresholds, the conflict detection module achieved an overall accuracy of 92.8% and a recall of 89.6%. This high level of detection accuracy provides a robust foundation for knowledge fusion, effectively preventing erroneous attribute merging and redundant expressions, and improving the semantic consistency and structural completeness of the fused knowledge graph.

3.3. Three-dimensional geological model construction

3.3.1. Construction of virtual boreholes

Both implicit and explicit three-dimensional modeling approaches rely on borehole data as essential spatial constraints. The textual and graphical information extracted in this study contains overlying and underlying relationships, lithostratigraphic vertical sequences, and thickness information, which together support the construction of virtual boreholes for spatial interpolation (Fig. 7).

During model construction, real geological cross-sections are

intentionally excluded from the virtual borehole generation and spatial interpolation processes, and are reserved solely for model validation after model completion. This design avoids the repeated use of identical data layers in both model construction and validation, while enabling a direct comparison between model-derived sections and real geological cross-sections, thereby allowing an independent and transparent assessment of stratigraphic sequence consistency and geometric reliability.

To evaluate the applicability of the method across different spatial scales, multi-resolution planar grids (100 m and 50 m) were generated from the vectorized geological main map (Fig. 8). The number of identified stratigraphic units was five, six, and seven for the 100-meter, 50-meter, and 10-meter resolutions respectively. As resolution increases, the model preserves more small-scale stratigraphic details. For example, the narrow formation unit K_1c^1 appears clearly only at the 10-meter resolution. These findings show that multi-scale rasterization effectively captures spatial detail in formation distribution and provides a hierarchical foundation for three-dimensional modeling.

3.3.2. Construction of the three-dimensional geological model using spatial interpolation

Using the entity-relation network, separate three-dimensional models were constructed for stratigraphic unit and geological ages. This study adopts a “global filling-conflict detection-local refinement” strategy, projecting virtual borehole sequences into three-dimensional space and performing spatial interpolation using the KNN algorithm with digital elevation model and boundary constraints. Models were built at three resolutions: 10 m, 50 m, and 100 m.

(1) Lithostratigraphic Model

The lithostratigraphic model emphasizes material composition and spatial contact relationships among geological bodies. As shown in Fig. 9, all three resolutions correctly reproduce the stratigraphic order and overall structural framework of the study area. Specifically, high resolution (10 m): The model surface is smooth and detailed, accurately

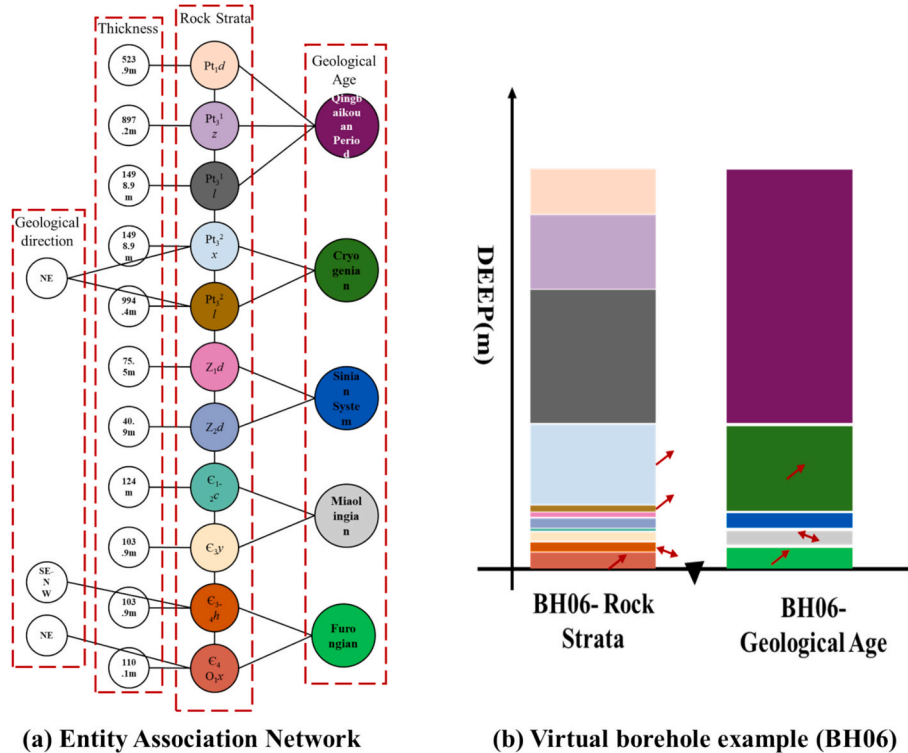


Fig. 7. Construction of virtual boreholes (Color represents lithostratigraphic or geological age categories; red arrows denote strike direction.).

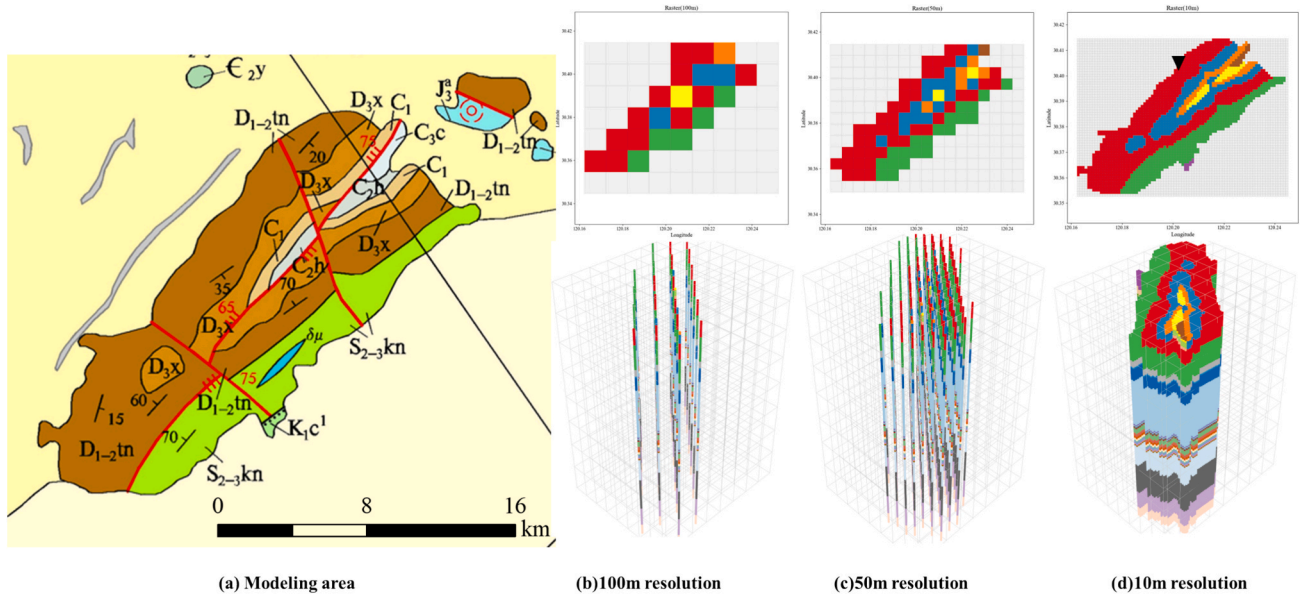


Fig. 8. Multi-scale rasterization of geological maps.

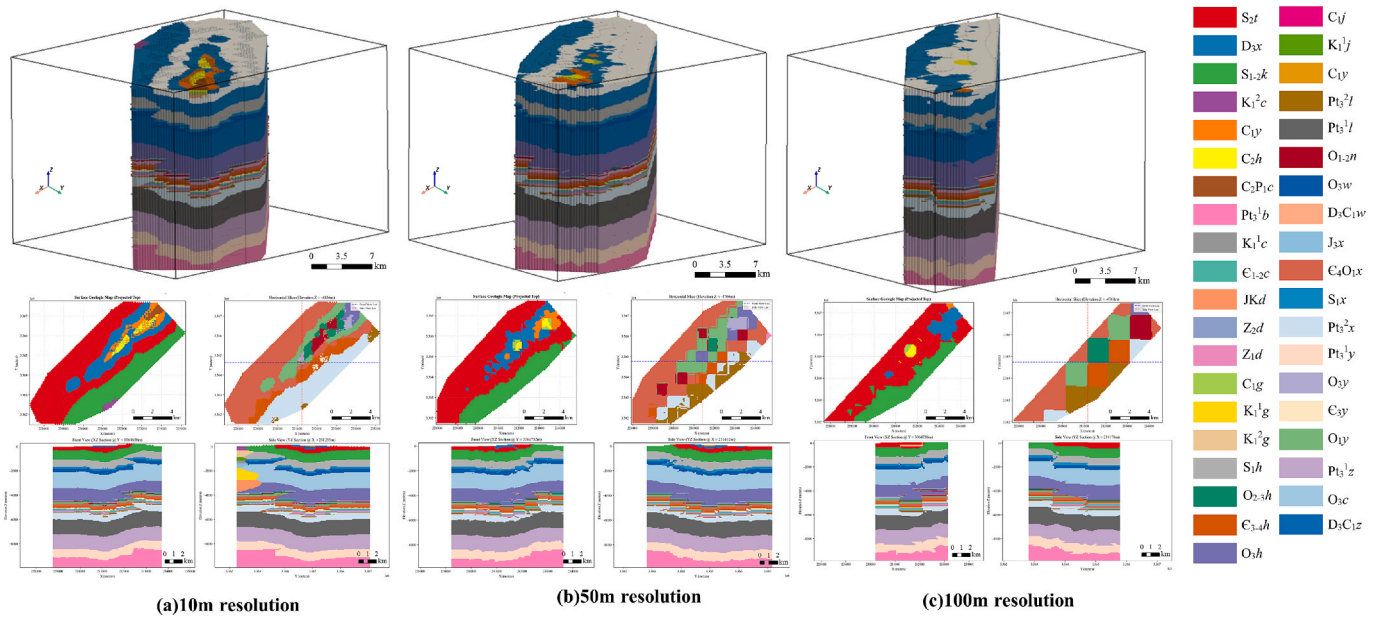


Fig. 9. Three-dimensional model and side view (stratigraphic unit).

depicting small-scale features such as lenses and pinch-out structures. It captures strike and dip changes across fault blocks with high sensitivity, making it suitable for fine-scale analysis in structurally complex areas. Medium resolution (50 m): Major geological boundaries remain clear, while minor lithological noise is smoothed out. The model offers a good balance between visualization efficiency and geological fidelity, making it ideal for regional to sub-regional structural analysis. Low resolution (100 m): The model shows an evident voxel-like step pattern with reduced detail. Nevertheless, it efficiently represents large-scale structural trends and suppresses local noise, making it useful for regional overviews and geological trend interpretation.

(2) Geological Age Model

The geological age model highlights the temporal evolution of geological bodies (Fig. 10). Formation units with identical geological

ages are automatically merged. Compared with the lithostratigraphic model, the age-based model suppresses lithological variation and emphasizes the spatial organization of geological time intervals. The model clearly illustrates the superposition sequence from older formations (e. g., Sinian and Cambrian) to younger deposits (e.g., Quaternary), revealing the sedimentary evolution of the region. As grid resolution decreases, unconformities and age boundaries become generalized. The 10-meter model sharply defines age contacts, while the 100-meter model captures broader age domains with more jagged boundaries but still preserves accurate temporal relationships in deeper structures.

3.3.3. Model construction efficiency and accuracy evaluation

This verification strategy follows the commonly used principles of input-output separation and consistency in geological modeling research, and relies on independent reference data [19,45,46]. Table 9 summarizes the computational indicators of the three-dimensional

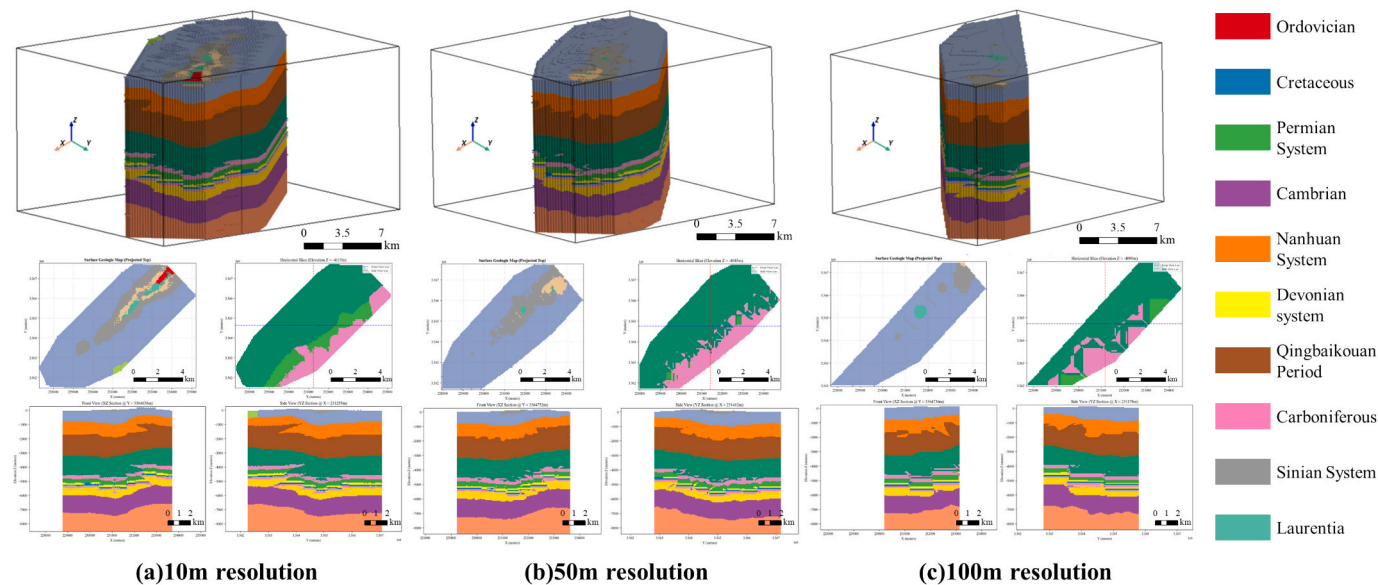


Fig. 10. Three-dimensional model and side view (geological age).

models. As the spatial resolution decreases from 10 m to 100 m, the construction time of the lithostratigraphic model reduces from 8.81 s to 2.92, indicating a substantial efficiency improvement. When conducting large-scale geological modeling, reducing the resolution can significantly save computational resources. At the same resolution, the geological age model requires slightly more computation time than the lithostratigraphic model. For example, at the 10-meter resolution, the computational cost is approximately 4.3% higher. This additional cost arises from the logical operations required for age merging and layer-category identification. Although it increases computational load slightly, it produces models with stronger geological consistency. The effective voxel volume decreases as resolution becomes coarser, reflecting the increased generalization of geological boundaries under low-resolution grid structures.

To validate the reliability of the constructed three-dimensional model, one representative borehole section (derived from standard geological cross-sections included in geological maps and reserved exclusively for independent validation after model construction) was compared with the corresponding model section at 10 m resolution. The evaluation focuses on two indicators: stratigraphic sequence consistency and thickness accuracy (Table 10). The modeled section successfully reconstructs the stratigraphic stacking sequence observed in the real geological section. From top to bottom, both sections show the sequence “Yejiatang Formation (YF)-Xihu Formation (YF)-Tangjiawu Formation (TF)-Kangshan Formation (KF),” with no stratigraphic inversion, omission, or incorrect intercalation. Thickness comparison shows that the model reproduces overall thickness trends, although discrepancies exist due to voxel resolution limits and the smoothing nature of the KNN interpolation. Local variations in true thickness, such as abrupt layer

transitions, tend to be smoothed during interpolation, leading to slight overestimation or underestimation (e.g., Kangshan Formation overestimated by 50 m).

The sequence consistency accuracy (ACC) for the section is 86.71%, demonstrating that the model retains high geometric fidelity even without deep drilling constraints, satisfying regional geological interpretation requirements.

4. Discussion

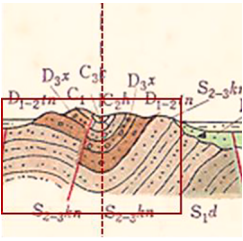
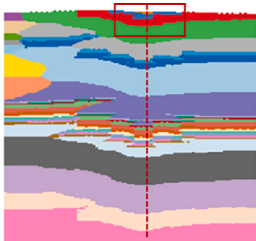
This study introduces a multimodal information fusion approach for three-dimensional geological modeling, addressing the challenge of subsurface characterization in regions with sparse drilling data. By jointly extracting planar geometric information from geological maps and semantic information from geological texts, a structured multimodal geological knowledge graph is constructed. Virtual boreholes are then generated to transform this discrete geological knowledge into continuous spatial constraints, including stratigraphic order, thickness, lithology, and attitude parameters. This establishes a new data paradigm for regional-scale voxel modeling.

Compared to traditional three-dimensional geological modeling methods, the construction cost of dense drilling data is high and its spatial distribution is uneven, thereby limiting its applicability at the regional scale [47–50]. The concept of “virtual boreholes” is an effective data augmentation method. However, in most existing studies, virtual boreholes are either generated through data-driven inference based on existing drilling samples (e.g., layer boundary and thickness prediction based on machine learning) or generated according to geological rules defined by experts (typically supported by engineering practice or geophysical inversion results) [19,51,52]. However, the performance of such methods based on virtual drilling is limited by the availability and representativeness of the drilling training data, as well as the expertise of the experts. When applied to regional-scale modeling or geological environments with strong heterogeneity and complex structures, uncertainties arising from numerical extrapolation or heuristic assumptions may increase, leading to reduced model stability away from the data control points. In contrast, the virtual drilling proposed in this study is generated through explicit geological reasoning based on the sequence of strata, contact relationships, and structural information extracted from geological maps and literature, rather than through numerical prediction or heuristic rule assignment. This knowledge-based mechanism endows the virtual drilling with clear geological semantics

Table 9
Overview of three-dimensional model construction.

Category	Resolution	Total Time	Total Voxels	Effective Geological Volume
Stratigraphic unit	10 m	8.81 s	4,139,565	2,095,529
	50 m	4.07 s	3,417,708	1,691,211
	100 m	2.92 s	3,094,236	1,144,515
Geological age	10 m	9.19 s	3,574,161	1,798,625
	50 m	5.87 s	2,929,464	1,449,499
	100 m	4.40 s	2,650,014	980,504

Table 10
Section consistency evaluation.

Real profile map	Modeling profile (10 m)	Rock strata – real	Thickness-True (m)	Rock strata-model	Thickness –model (m)
		YF	53	YF	66
		YF	118	YF	121
		TF	186	TF	210
		KF	320	KF	370

and traceable derivation paths, enhancing the interpretability of the model and enabling systematic updates as new geological information becomes available. From the perspective of the modeling results, the existing methods that rely on real drilling data usually achieve relatively high local accuracy, especially in the vicinity of the real drilling points or within the distribution range of the training samples. However, their reliability may decrease in areas far from the data control points or in regions with complex structural conditions, where the uncertainty of extrapolation becomes the dominant factor [53–55]. In contrast, the proposed framework emphasizes regional-scale structural consistency by enforcing the stratigraphic sequence and topological relationships derived from geological knowledge. Combined with multi-resolution rasterization technology, this framework captures large-scale structural patterns at a coarse resolution while retaining local stratigraphic features at a finer scale, thereby supporting coherent regional modeling and engineering-oriented applications.

The three-dimensional geological model construction method proposed in this paper can be used as an effective preliminary modeling tool in practical applications, providing support for regional geological surveys, engineering planning, and disaster assessment, especially in the early investigation stages where drilling resources are limited. Moreover, the knowledge-driven virtual boreholes generated by this method can serve as a decision support layer, helping to identify areas with high geological uncertainty, thereby facilitating the prioritization and optimization of subsequent drilling activities. This method does not replace detailed site-scale surveys, but is most suitable as a supplementary framework to fill the gap between surface geological information and underground modeling, providing an interpretable and updatable geological background for further data collection and fine modeling.

This study constructs a three-dimensional geological model by integrating multimodal geological information from maps and texts, mainly focusing on geological map information and geological textual information. However, geological geophysical data (such as seismic, gravity, or electrical data) have not been included yet, which may limit the model's ability to interpret deep-buried or concealed geological structures. Secondly, the current depth of exploration of geological map information is insufficient, which may lead to incomplete or inaccurate information extraction. Thirdly, the spatial interpolation strategy adopted retains the discreteness of strata and regional structural trends, but has limited ability to represent extremely complex geological features (such as overturned folds or concealed fault systems with strong nonlinearity or discontinuity). Future research will integrate relevant multimodal information to conduct deeper exploration of geological maps under professional domain guidance, in order to improve the accuracy and modeling capabilities of the model.

5. Conclusion

This study presents a regional-scale three-dimensional stratigraphic modeling framework that integrates geological maps and texts to address the challenge of subsurface modeling in borehole-scarce

environments. By constructing a multi-modal knowledge graph that fuses geometric features from maps with semantic information from geological descriptions, abstract geological knowledge is transformed into explicit spatial constraints in the form of virtual boreholes with clear geological semantics and traceable derivation paths. Coupled with a multi-resolution rasterization and interpolation strategy, the proposed approach enables the construction of three-dimensional geological models that balance geological detail and computational efficiency. Validation results demonstrate that the resulting lithostratigraphic model achieves a stratigraphic sequence consistency accuracy of 86.71%, indicating that the framework can maintain geological coherence and structural reliability even under data-limited conditions. Overall, this study provides a scalable and interpretable solution for regional three-dimensional geological modeling and offers a novel pathway for converting heterogeneous geological knowledge into engineering-ready subsurface models.

Despite these advantages, several limitations and future research directions remain. First, the current framework primarily relies on two-dimensional geological maps and textual reports; the integration of additional data sources such as geophysical profiles, remote sensing constraints, and sparse real borehole measurements could further enhance model robustness and resolution. Second, although virtual boreholes provide effective vertical constraints, their uncertainty propagation and quantitative impact on model reliability warrant more systematic investigation. Future work will also focus on improving automated knowledge extraction, incorporating uncertainty-aware modeling strategies, and extending the framework toward dynamic geological processes and application-driven scenarios such as ground-water simulation, resource assessment, and geohazard analysis.

CRediT authorship contribution statement

Qirui Wu: Writing – review & editing, Writing – original draft, Methodology, Data curation. **Zhong Xie:** Writing – review & editing, Methodology. **Yifan Zhao:** Writing – review & editing. **Miao Tian:** Validation. **Xiaotian Zhang:** Writing – review & editing. **Qinjun Qiu:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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